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In [6]: import numpy as np
import pandas as pd
import torch
import torch.nn as nn
import torch.optim as optim
from sklearn.metrics import (
    f1_score, recall_score, precision_score, accuracy_score, roc_auc_score,
    confusion_matrix, classification_report, average_precision_score
)
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
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In [7]: %pwd
```

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Out[7]: 'C:\\Users\\anton\\Documents\\programming\\python\\ML\\QNN'
```

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In [8]: # =====
# ♦ Dataset
# =====
df = pd.read_csv("../creditcard.csv")

X = df.drop(columns=["Class"]).values
y = df["Class"].values

scaler = StandardScaler()
X = scaler.fit_transform(X)

# Separando índices de fraude e não fraude
fraudes = np.where(y == 1)[0]
nao_fraudes = np.where(y == 0)[0]
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In [9]: # Amostras para treino
train_fraudes = np.random.choice(fraudes, 390, replace=False)
train_nao_fraudes = np.random.choice(nao_fraudes, 1000, replace=False)

# Amostras para teste
test_nao_fraudes = np.random.choice(
    np.setdiff1d(nao_fraudes, train_nao_fraudes), 1000, replace=False
)
test_fraudes = np.setdiff1d(fraudes, train_fraudes)

train_idx = np.concatenate([train_fraudes, train_nao_fraudes])
test_idx = np.concatenate([test_fraudes, test_nao_fraudes])

X_train, y_train = X[train_idx], y[train_idx]
X_test, y_test = X[test_idx], y[test_idx]

# Convertendo para tensores
X_train = torch.tensor(X_train, dtype=torch.float32).unsqueeze(1).to(DEVICE) #
(batch, seq=1, features)
y_train = torch.tensor(y_train, dtype=torch.float32).unsqueeze(1).to(DEVICE)

X_test = torch.tensor(X_test, dtype=torch.float32).unsqueeze(1).to(DEVICE)
y_test = torch.tensor(y_test, dtype=torch.float32).unsqueeze(1).to(DEVICE)
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In [19]: # =====
#   ♦ LSTM "by hand"
#   =====
class MyLSTM(nn.Module):
    def __init__(self, input_size, hidden_size, output_size=1):
        super(MyLSTM, self).__init__()
        self.hidden_size = hidden_size

        # Pesos
        self.W_i = nn.Linear(input_size + hidden_size, hidden_size)
        self.W_f = nn.Linear(input_size + hidden_size, hidden_size)
        self.W_c = nn.Linear(input_size + hidden_size, hidden_size)
        self.W_o = nn.Linear(input_size + hidden_size, hidden_size)

        self.fc = nn.Linear(hidden_size, output_size)
        self.sigmoid = nn.Sigmoid()

    def forward(self, x):
        h_t = torch.zeros(x.size(0), self.hidden_size).to(DEVICE)
        c_t = torch.zeros(x.size(0), self.hidden_size).to(DEVICE)

        for t in range(x.size(1)): # passo temporal
            x_t = x[:, t, :]
            combined = torch.cat((x_t, h_t), dim=1)

            i_t = torch.sigmoid(self.W_i(combined))
            f_t = torch.sigmoid(self.W_f(combined))
            g_t = torch.tanh(self.W_c(combined))
            o_t = torch.sigmoid(self.W_o(combined))

            c_t = f_t * c_t + i_t * g_t
            h_t = o_t * torch.tanh(c_t)

        out = self.fc(h_t)
        return self.sigmoid(out)

```

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In [21]: # =====
#   ♦ Treinamento
#   =====
model = MyLSTM(input_size=X.shape[1], hidden_size=32).to(DEVICE)
criterion = nn.BCELoss()
optimizer = optim.Adam(model.parameters(), lr=1e-3)

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In [22]: v_loss = []
EPOCHS = 500
for epoch in range(EPOCHS):
    model.train()
    optimizer.zero_grad()
    preds = model(X_train)
    loss = criterion(preds, y_train)
    loss.backward()
    optimizer.step()
    v_loss.append(loss.item())
    print(f"Epoch {epoch+1}/{EPOCHS}, Loss: {loss.item():.4f}")

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Epoch 1/500, Loss: 0.7404
Epoch 2/500, Loss: 0.7352
Epoch 3/500, Loss: 0.7302
Epoch 4/500, Loss: 0.7253
Epoch 5/500, Loss: 0.7205
Epoch 6/500, Loss: 0.7157
Epoch 7/500, Loss: 0.7111
Epoch 8/500, Loss: 0.7065
Epoch 9/500, Loss: 0.7020
Epoch 10/500, Loss: 0.6975
Epoch 11/500, Loss: 0.6931
Epoch 12/500, Loss: 0.6887
Epoch 13/500, Loss: 0.6845
Epoch 14/500, Loss: 0.6802
Epoch 15/500, Loss: 0.6761
Epoch 16/500, Loss: 0.6720
Epoch 17/500, Loss: 0.6679
Epoch 18/500, Loss: 0.6639
Epoch 19/500, Loss: 0.6600
Epoch 20/500, Loss: 0.6562
Epoch 21/500, Loss: 0.6523
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Epoch 23/500, Loss: 0.6449
Epoch 24/500, Loss: 0.6413
Epoch 25/500, Loss: 0.6377
Epoch 26/500, Loss: 0.6342
Epoch 27/500, Loss: 0.6307
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Epoch 29/500, Loss: 0.6240
Epoch 30/500, Loss: 0.6207
Epoch 31/500, Loss: 0.6175
Epoch 32/500, Loss: 0.6143
Epoch 33/500, Loss: 0.6111
Epoch 34/500, Loss: 0.6080
Epoch 35/500, Loss: 0.6050
Epoch 36/500, Loss: 0.6019
Epoch 37/500, Loss: 0.5990
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Epoch 39/500, Loss: 0.5931
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Epoch 41/500, Loss: 0.5874
Epoch 42/500, Loss: 0.5846
Epoch 43/500, Loss: 0.5818
Epoch 44/500, Loss: 0.5790
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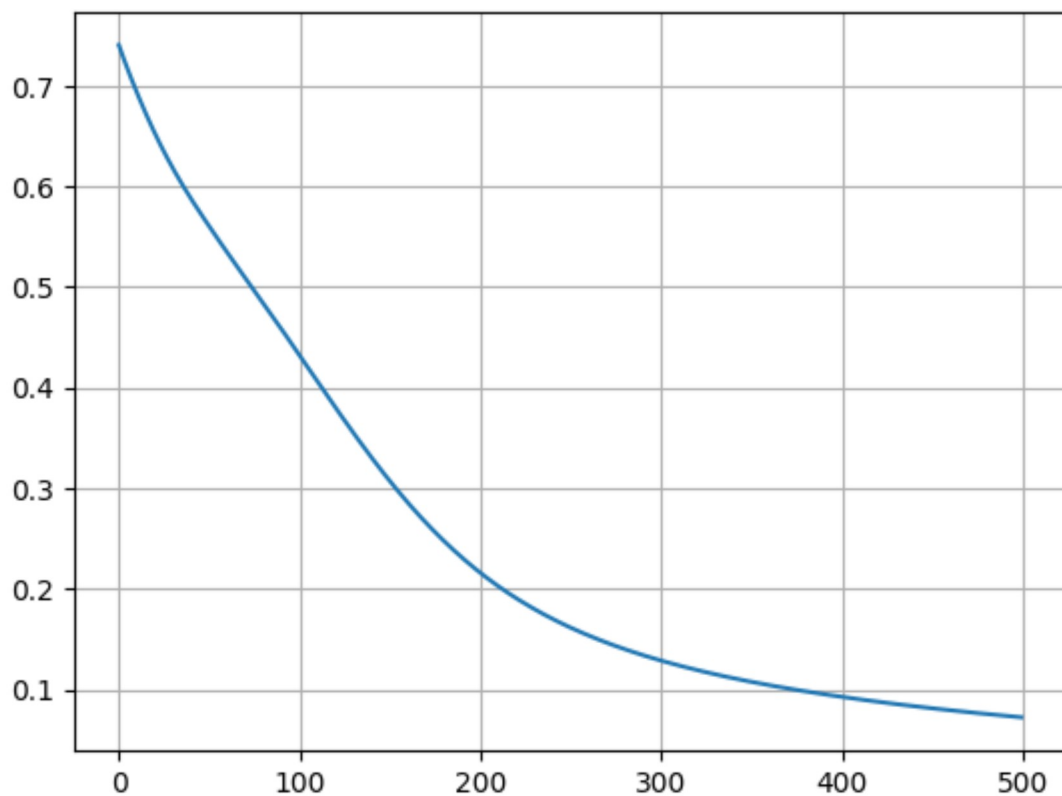
Epoch 361/500, Loss: 0.1040
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Epoch 474/500, Loss: 0.0767
Epoch 475/500, Loss: 0.0766
Epoch 476/500, Loss: 0.0764
Epoch 477/500, Loss: 0.0762
Epoch 478/500, Loss: 0.0760
Epoch 479/500, Loss: 0.0758
Epoch 480/500, Loss: 0.0757

```
Epoch 481/500, Loss: 0.0755  
Epoch 482/500, Loss: 0.0753  
Epoch 483/500, Loss: 0.0751  
Epoch 484/500, Loss: 0.0750  
Epoch 485/500, Loss: 0.0748  
Epoch 486/500, Loss: 0.0746  
Epoch 487/500, Loss: 0.0744  
Epoch 488/500, Loss: 0.0743  
Epoch 489/500, Loss: 0.0741  
Epoch 490/500, Loss: 0.0739  
Epoch 491/500, Loss: 0.0737  
Epoch 492/500, Loss: 0.0736  
Epoch 493/500, Loss: 0.0734  
Epoch 494/500, Loss: 0.0732  
Epoch 495/500, Loss: 0.0731  
Epoch 496/500, Loss: 0.0729  
Epoch 497/500, Loss: 0.0727  
Epoch 498/500, Loss: 0.0726  
Epoch 499/500, Loss: 0.0724  
Epoch 500/500, Loss: 0.0722
```

```
In [23]: import matplotlib.pyplot as plt
```

```
In [24]: plt.plot(v_loss)  
plt.grid()  
plt.show()
```



```
In [25]: # =====
#  ♦ Avaliação
# =====
model.eval()
with torch.no_grad():
    y_pred_prob = model(X_test).cpu().numpy().flatten()
    y_pred = (y_pred_prob >= 0.5).astype(int)
    y_true = y_test.cpu().numpy().flatten()
```

```
In [26]: # Métricas
f1 = f1_score(y_true, y_pred, average="macro")
recall = recall_score(y_true, y_pred, average="macro")
precision = precision_score(y_true, y_pred, average="macro")
accuracy = accuracy_score(y_true, y_pred)
auc = roc_auc_score(y_true, y_pred_prob)
auc_pr = average_precision_score(y_true, y_pred_prob)

# Confusion matrix
cm = confusion_matrix(y_true, y_pred)
tn, fp, fn, tp = cm.ravel()
frr = fn / (fn + tp) if (fn + tp) > 0 else 0

print("\n=== Resultados ===")
print(f"F1-score (macro): {f1:.4f}")
print(f"Recall (macro): {recall:.4f}")
print(f"Precision (macro): {precision:.4f}")
print(f"Accuracy: {accuracy:.4f}")
print(f"AUC-ROC: {auc:.4f}")
print(f"AUC-PR: {auc_pr:.4f}")
print(f"FRR: {frr:.4f}")
print("\nConfusion Matrix:\n", cm)
print("\nClassification Report:\n", classification_report(y_true, y_pred,
digits=4))
```

```
=== Resultados ===
F1-score (macro): 0.9310
Recall (macro): 0.9386
Precision (macro): 0.9237
Accuracy: 0.9764
AUC-ROC: 0.9620
AUC-PR: 0.9418
FRR: 0.1078
```

```
Confusion Matrix:
[[985  15]
 [ 11  91]]
```

```
Classification Report:
              precision    recall  f1-score   support

     0.0       0.9890      0.9850      0.9870        1000
     1.0       0.8585      0.8922      0.8750         102

   accuracy              0.9764        1102
  macro avg              0.9237      0.9386      0.9310        1102
 weighted avg              0.9769      0.9764      0.9766        1102
```

```
In [ ]: === Resultados ===
F1-score (macro): 0.9048
```

```
Recall (macro): 0.9370
Precision (macro): 0.8782
Accuracy: 0.9655
AUC-ROC: 0.9650
AUC-PR: 0.9386
FRR: 0.0980
```

Confusion Matrix:

```
[[972  28]
 [ 10  92]]
```

Classification Report:

	precision	recall	f1-score	support
0.0	0.9898	0.9720	0.9808	1000
1.0	0.7667	0.9020	0.8288	102
accuracy			0.9655	1102
macro avg	0.8782	0.9370	0.9048	1102
weighted avg	0.9692	0.9655	0.9668	1102